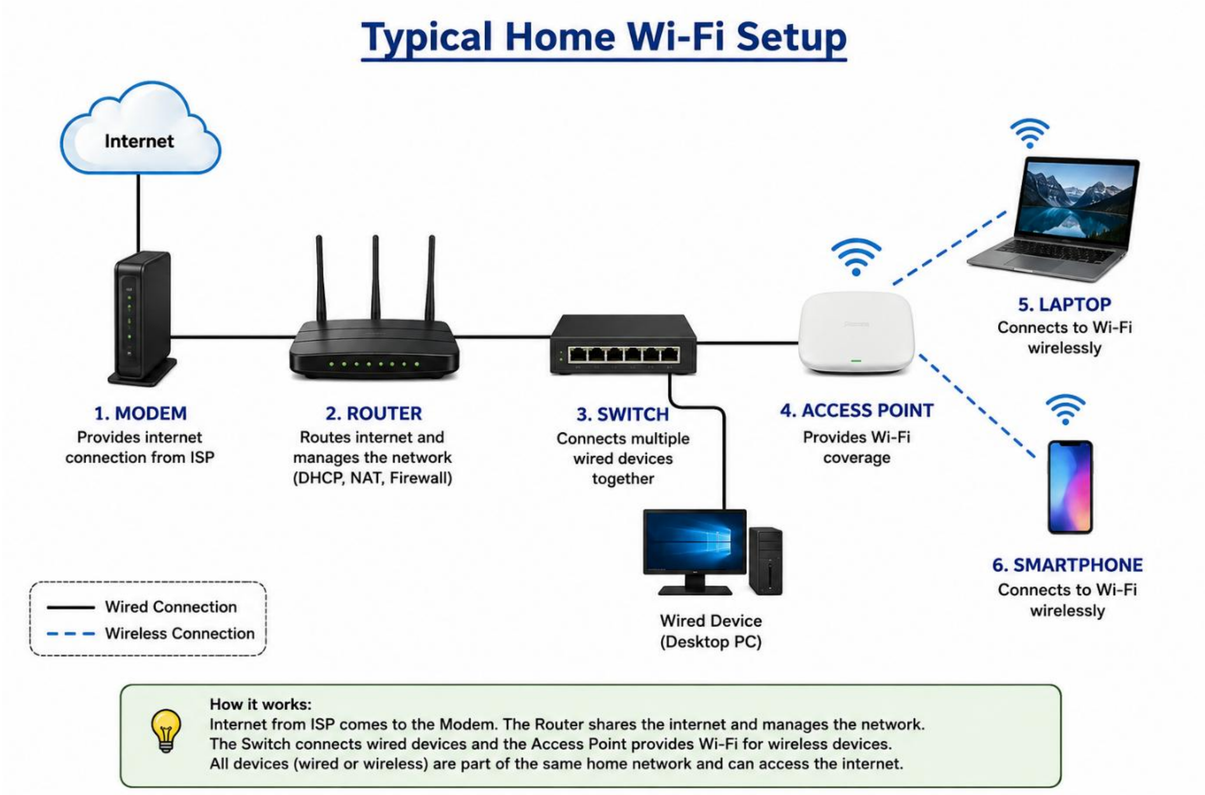
A+ : UNDERSTANDING OF NETWORK

NETWORK

TASK: 1

Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.



The above diagram shows a typical home Wi-Fi network setup. The internet connection first enters the **modem** from the Internet Service Provider (ISP). The modem passes the connection to the **router**, which manages the network and shares the internet connection with multiple devices. A **switch** can be connected to the router to provide additional wired connections for devices such as desktop computers, printers, or smart TVs. The **access point** provides Wi-Fi coverage, allowing wireless devices such as laptops and smartphones to connect to the network. Wired connections are usually made using Ethernet cables, while devices like smartphones and laptops can connect wirelessly through Wi-Fi. Together, these devices create a complete home network that allows users to access the internet and communicate between connected devices.

TASK: 2

List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

Networking Device	Main Function	Real-Life Example
Modem	Connects a home or office network to the Internet Service Provider (ISP) and converts the internet signal into a usable network connection.	**At home**, the modem receives the internet connection from the ISP.
Router	Directs data between devices and networks and allows multiple devices to share one internet connection.	**At home**, a router shares Wi-Fi with laptops, smartphones, and smart TVs.
Switch	Connects multiple wired devices together on the same local network.	**In an office**, a switch connects multiple computers, printers, and servers using Ethernet cables.
Access Point (AP)	Provides wireless Wi-Fi access to devices and extends wireless network coverage.	**In a large office or college**, access points provide Wi-Fi in different rooms and areas.
Network Cable (Ethernet Cable)	Transfers data between networking devices through a physical wired connection.	**At home or in an office**, an Ethernet cable connects a computer to a router or switch.

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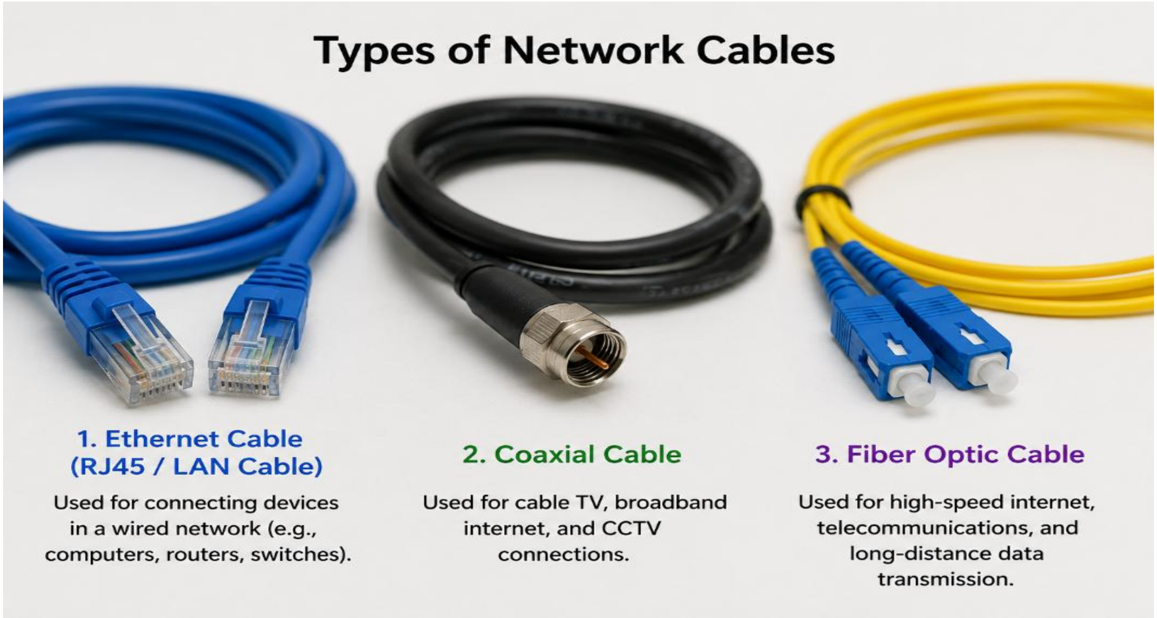
TASK: 3

Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.



1. Ethernet Cable (RJ45 / LAN Cable)

Used for connecting devices such as computers, routers, switches, and printers in a wired network.

2. Coaxial Cable

Commonly used for cable TV connections, broadband internet, and CCTV systems.

3. Fiber Optic Cable

Used for high-speed internet and long-distance data transmission.

TASK: 4

Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

Feature	Router	Switch
Main Use	Connects different networks and provides internet access.	Connects multiple devices within the same local network (LAN).
Typical Location	Commonly used at homes, offices, schools, and businesses near the internet connection.	Commonly used in offices, schools, computer labs, and large networks with many wired devices.
How It Handles Data	Examines IP addresses and forwards data between different networks.	Examines MAC addresses and sends data to the correct device within the same network.
Internet Sharing	Shares one internet connection with multiple devices.	Does not directly provide internet; it distributes network connections between wired devices.
Wireless Support	Many routers provide Wi-Fi connectivity.	Most standard switches provide only wired Ethernet connections.
Network Layer	Mainly works at the **Network Layer (Layer 3)** of the OSI model.	Mainly works at the **Data Link Layer (Layer 2)** of the OSI model.

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